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Smith Display : The Smith Chart

The Smith Chart

The Smith Display plots scattering parameter (s-parameter) data on a Smith chart. A Smith chart is a polar plot of the reflection coefficient Γ superimposed on a background of orthogonal circles. The Smith chart has two background styles: impedance and admittance.

The magnitude, or modulus, of plotted Γ is the distance from the center of the chart, and the phase angle of Γ is the angle around the chart from the positive x axis. Typically, all values on the impedance chart are normalized to a reference or characteristic impedance Z_0 (usually 50 ohms), so that the outermost circle of the chart represents Γ=1 (R=0), and the center of the chart is at Z_0.

The background circles and arcs are drawn inside the outermost circle (Γ=1), which is marked off counterclockwise in increments of 10 degrees from 0 to 180 outside the top half of the circle, and from -10 to -180, clockwise, outside the bottom half. Note that the only labels inside the outer circle are the values of the inner circles and arcs. That is, the vertical axis does not appear on the chart, and the horizontal axis is not labeled with any ticks or numbers indicating x values. Rather, it is labeled with the values of the circles where they intersect the horizontal axis. The arcs are labeled where they intersect the outermost circle of the chart. Additional labels of arc and circle values may be used at their intersections, depending on the density of the circles and arcs shown on the chart.

When the Smith chart background is set to impedance, the circles centered on the horizontal axis are the constant resistance (R) circles, and the arcs of circles centered on the vertical axis are the constant reactance (X) arcs. These circles and arcs are labeled with the values of constant R and X, respectively, that they represent. The constant resistance circles are plotted (in rectangular coordinates) as follows: the center of each circle is at x= R/(R+1), y=0, and the radius = 1/(R+1). The constant reactance arcs are plotted with centers at x=1, y=1/X, and radius 1/X.

The Smith chart allows a quick graphical translation between the reflection coefficient Γ and impedance Z, while limiting their magnitudes to be between 0 and 1 to make for easier plotting. Any point plotted on the impedance background will always be at the intersection of some circle (R) and some arc (X), and so represents a complex series impedance combination Z = R + jX. The magnitude and phase angle of the plotted point, if not already supplied in that form, can be easily obtained if the point is given in terms of rectangular coordinates.

The Smith chart's admittance background is used for analysis of parallel combinations of RF circuit elements and is simply the impedance background rotated 180 degrees. Circles on the horizontal axis are now the circles of constant conductance (G), and arcs of the circles centered on the vertical axis are now the constant susceptance (B) circles, and they are labeled with their values accordingly. Any point plotted on the admittance background is at the intersection of some circle (G) and some arc (B) and so represents a series admittance Y = G + jB, which is used as a more convenient way of representing parallel impedances. Thus, in the same manner as the impedance background is used, G may be easily translated to admittance, and vice-versa, using the admittance background of the Smith chart. The admittance chart is usually normalized by $1/Z_0$.

For the implementation of the Smith chart in the Smith Display, see S1P Smith Display.

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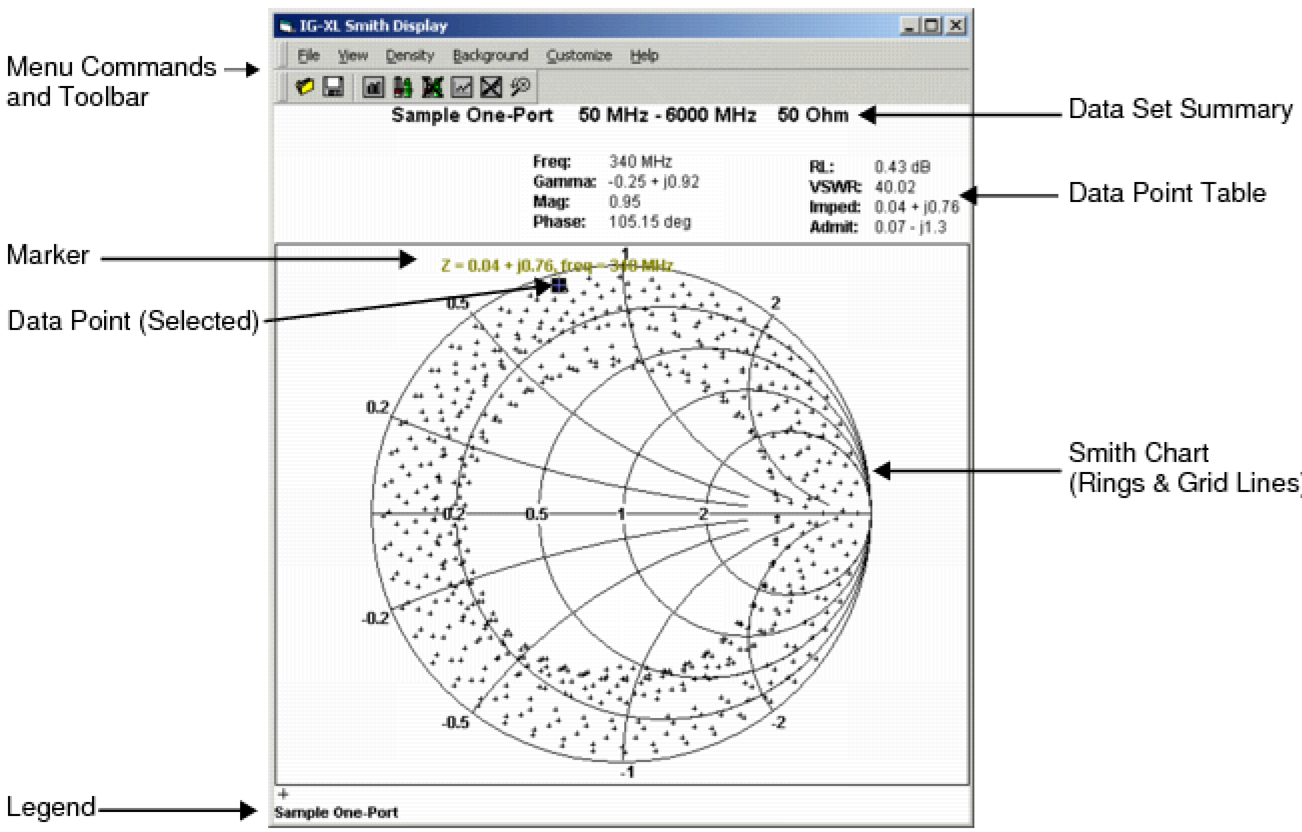
Smith Display : S1P Smith Display

S1P Smith Display

The S1P Smith Display plots s-parameter data on a Smith chart. It is most useful for plotting forward and reverse reflection impedances, s11 and s22. See the figure.



One-Port Smith Display

The display includes a Smith chart polar plot grid with all the data points in a loaded data set. When you click a particular data point, the tool shows information about that point in the Data Point Table above the Smith chart. Similarly, when you turn on markers and click a data point, the Smith Display displays a text string near the data point with complex impedance and frequency values.

For more information, see Smith Display Commands and Toolbars.

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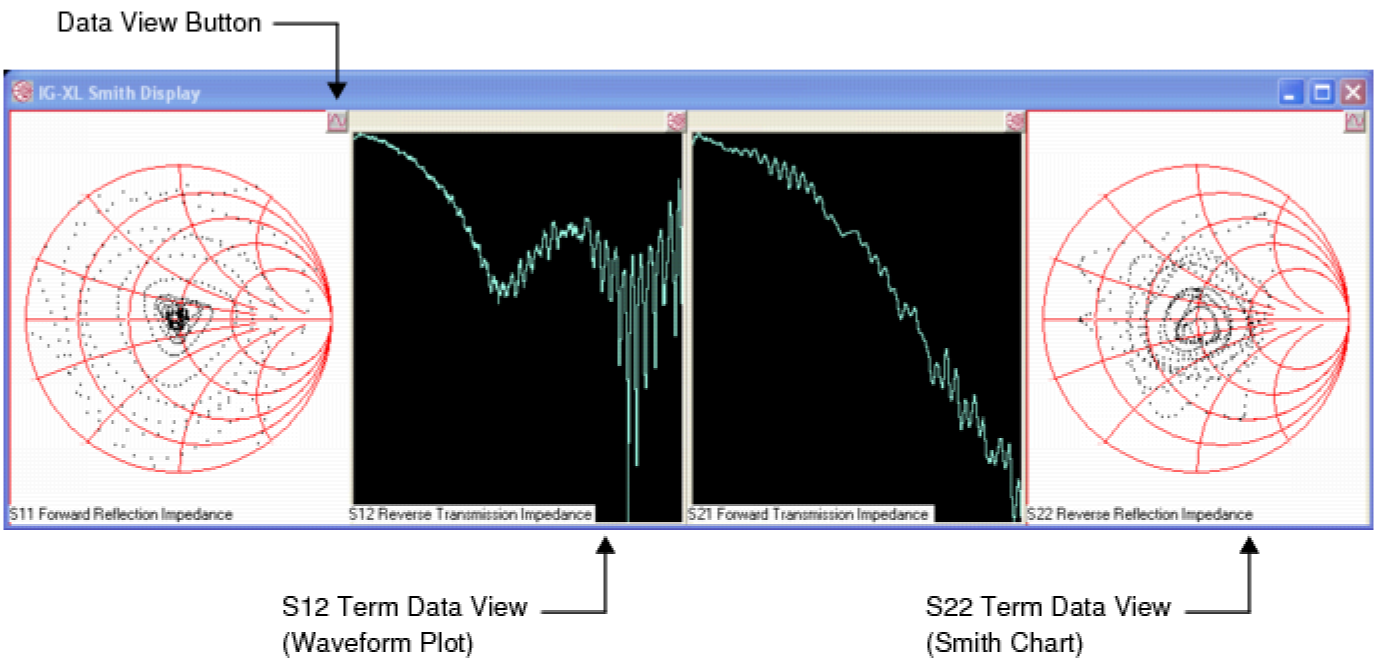
Smith Display : S2P Smith Display

S2P Smith Display

The S2P Smith Display plots data sets for all four two-port s-parameter terms in a single window. The window has four compact plot displays (data view panes), one for each s-parameter term. See the figure.



S2P Smith Display Window

Each data view pane can display a Smith chart or a waveform plot. By default, the S2P Smith display opens with the following data views:

- Smith charts for s11 and s22, the reflection impedance terms
- Waveform plots for s12 and s21, the transmission impedance terms

S2P Smith Display Commands

The S2P Smith Display allows you to:

- Change back and forth from the Smith chart view to the waveform view, in each pane
- Launch a full-featured display tool (Smith Display or WaveScope) for detailed analysis

To change the data plot shown in a single data view pane:

- Click the data view button at the upper right corner of the pane. The button appears as a sine wave if the Smith chart is displayed or a Smith chart if the waveform is displayed.

The data view changes from a Smith chart to a waveform or from a waveform to a Smith chart. The button also changes.

To launch an S1P Smith Display for a Smith chart or a WaveScope for a waveform:

- Double-click the data view pane for the s-parameter term you want to analyze.

A S1P Smith Display window or WaveScope window opens with the data set loaded.

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Smith Display : Smith Display Commands and Toolbars

Smith Display Commands and Toolbars

This section includes the following topics:

- Menus and Toolbar Icons
- Data Sets Dialog Box
- Customization Dialog Box (General Tab)
- Customization Dialog Box (Plot Tab)
- Customization Dialog Box (Font Tab)
- Customization Dialog Box (Color Tab)
- Customization Dialog Box (Style Tab)
- Export Dialog Box

For S2P Smith Display commands, see S2P Smith Display.

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Smith Display : Smith Display Commands and Toolbars : Menus and Toolbar Icons

Menus and Toolbar Icons

The following topics describe the S1P Smith Display menus and toolbar commands:

- File Menu Commands
- View Menu Commands
- Density Menu Commands
- Background Menu Commands
- Customize Menu Commands

For S2P Smith Display commands, see S2P Smith Display.

File Menu Commands

Open	 (Ctrl+O) Opens a file containing the s-parameter data set to plot on the Smith Chart. See Loading Data Sets Using the Open Command.
Save	 (Ctrl+S) Saves one or more plotted data sets to a file. See Saving Data Sets and Plots.

View Menu Commands

Data Sets	 Opens the S-Parameter Data Sets dialog box, which allows you to show or hide selected s-parameter data sets and to set the frequency range for a selected data set. See: • Data Sets Dialog Box • Displaying Data Point Information Using the Data Sets Command
Markers On	 Activates the marker feature. Markers are strings of text that show the frequency associated with a data point and the associated impedance Z or admittance Y, according to the selected chart background. Markers are turned off by default. See Displaying Data Point Information Using Markers.
Markers Off	 Deactivates the marker feature. Markers are turned off by default. See Displaying Data Point Information Using Markers.
Show Data On	 Activates the Data Point Table that displays detailed data for any plotted point you click. Show Data On is active by default and can be deactivated using the Show Data Off button. The Data Point Table displays the selected data point's frequency, gamma, magnitude, phase, return loss (RL), voltage standing wave ratio (VSWR), impedance, and admittance. See Displaying Data Point Information in the Data Point Table.
Show Data Off	 Deactivates the Data Point Table. Show Data is active by default. When deactivated, the entire Data Point Table disappears from the display. Use the Show Data On button to have the table reappear in the display. See Displaying Data Point Information in the Data Point Table.
Show Legend	Displays a legend at the bottom of the display that identifies the data point symbol used on the Smith Chart to differentiate data points of one

	data set from another. By default, the legend is hidden. See Showing and Hiding the Smith Display Legend.
Legend Off	Hides the legend at the bottom of the display. The legend is hidden unless the Show Legend button is used. See Showing and Hiding the Smith Display Legend.
Undo Zoom	 Returns the display to its original size and viewing state after a zoom action has been performed. See Zooming In and Out of a Data Point on the Smith Display.

Density Menu Commands

Controls the number of constant resistance (R) circles and constant reactance (X) arcs (or G and B) rings that appear on the Smith Chart. Three ring densities are available:

- Simple
- Intermediate (default)
- Dense

For more information, see Changing the Ring Density.

☑ **Note:**

Changing ring densities does not affect plotted data or markers.

Background Menu Commands

Alternates the Smith Chart background between the impedance and admittance backgrounds. The impedance background is the default.

For more information, see Changing the Background Between Impedance and Admittance.

☑ **Note:**

Switching backgrounds does not affect plotted data or markers. Also, the impedance and admittance backgrounds cannot be displayed simultaneously.

Customize Menu Commands

Customization Dialog	Opens a multitab dialog box that allows you to customize the appearance of the display and plotted data. Among the settings that you can change are the color and plotting style of the data points, background color, fonts, and Smith Chart grid lines.

	Note that when the Customization dialog box is open, the control settings reflect the current state of the display. See: • Customization Dialog Box (General Tab) • Customization Dialog Box (Plot Tab) • Customization Dialog Box (Font Tab) • Customization Dialog Box (Color Tab) • Customization Dialog Box (Style Tab)
Export Dialog	Opens a dialog box that allows you to export an image of the display to the clipboard, a file, or to a printer. See: • Export Dialog Box • Saving Data Sets and Plots • Printing a Smith Display Plot

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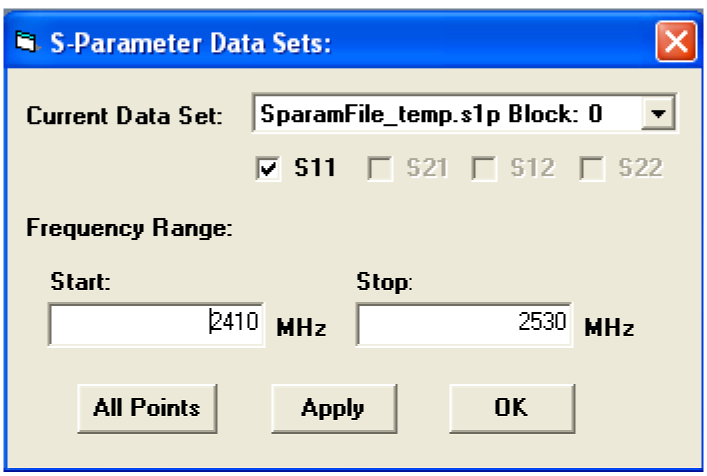
Smith Display : Smith Display Commands and Toolbars : Data Sets Dialog Box

Data Sets Dialog Box

The View>Data Sets command opens the S-Parameter Data Sets dialog box, which allows you to show or hide selected s-parameter data sets and to set the frequency range for the selected data set.



S-Parameter Data Sets Dialog Box

Current Data Set	A drop-down list that contains the names of all data sets currently loaded into the display. The settings specified on the S-parameter dialog box apply to the selected data set.
S11	Shows or hides the selected S11 trace from the display. Deselect this box to remove the selected S11 trace from the display. To restore the trace, click the box again to activate it. See Displaying Data Point Information Using the Data Sets Command.
S21	Two-port data only. Shows or hides the selected S21 trace from the display. Deselect this box to remove the selected S21 trace from the

	display. To restore the trace, click the box again to activate it. See Displaying Data Point Information Using the Data Sets Command.
S12	Two-port data only. Shows or hides the selected S12 trace from the display. Deselect this box to remove the selected S12 trace from the display. To restore the trace, click the box again to activate it. See Displaying Data Point Information Using the Data Sets Command.
S22	Two-port data only. Shows or hides the selected S22 trace from the display. Deselect this box to remove the selected S22 trace from the display. To restore the trace, click the box again to activate it. See Displaying Data Point Information Using the Data Sets Command.
Start	Sets the frequency at which the data set starts plotting on the Smith Chart. Use this setting to limit the data plotted on the chart. The default value is the first frequency in the data set.
Stop	Sets the frequency at which the data set stops plotting data on the Smith Chart. Use this setting to limit the data plotted on the chart. The default value is the last frequency in the data set.
All Points	Restores the default values for the start and stop frequency.
Apply	Previews plotting of the selected data set from the specified start frequency to the specified stop frequency. You must click OK to complete any changes.
OK	Plots the selected data set from the specified start frequency to the specified stop frequency and dismisses the Data Sets window.

For more information, see Displaying Data Point Information Using the Data Sets Command.

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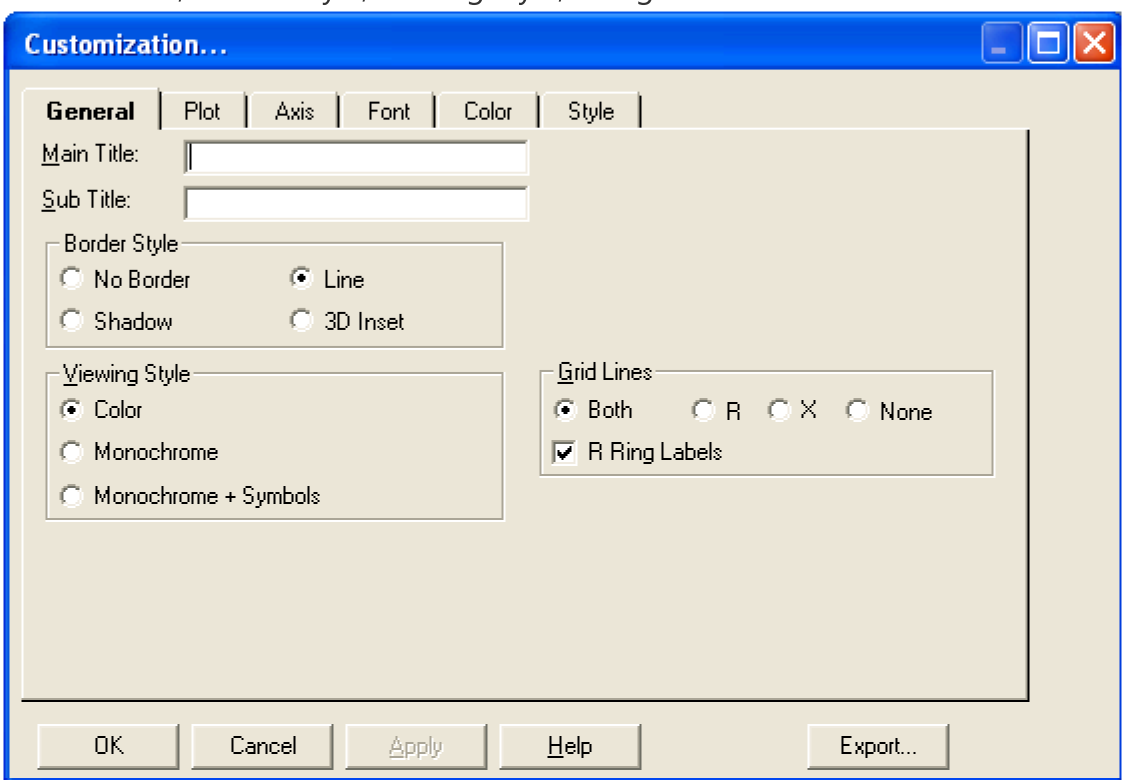
Smith Display : Smith Display Commands and Toolbars : Customization Dialog Box (General Tab)

Customization Dialog Box (General Tab)

The Customize>Customization Dialog command opens the multitabbed Customization dialog box that allows you to change the appearance of the Smith Display and plotted data.

The General tab allows you to change some visual attributes of the display window including the data set title, border style, viewing style, and grid lines.



General Tab Controls

Main Title	This setting is not implemented and does not affect the display.
Sub Title	This setting is not implemented and does not affect the display.

Border Style	Activates and specifies the style of the border that frames the Smith Chart in the display. Options: • No Border • Line • Shadow • 3D Inset
Viewing Style	Adjusts the display's image for printing on a monochrome printer. If you are including fewer than four data sets in a graph, use the Monochrome setting. If you are including more than four data sets, use Monochrome + Symbols to help distinguish different subsets. Options: • Color • Monochrome • Monochrome + Symbols See Printing a Smith Display Plot.
Grid Lines	Shows or hides the vertical (X) and horizontal (R) grid lines that make up the Smith Chart. The Smith Chart can contain vertical grid lines, horizontal grid lines, both vertical and horizontal grid lines, or no grid lines. • Both: Shows both the R and X rings. • R: Shows the R rings only. Hides the X rings. • X: Shows the X rings only. Hides the R rings.

- None: Shows no grid lines at all, nor the center horizontal line.

- R Ring Labels: Shows/hides the reference labels along the R ring. Note that R rings must be active (R or Both) for R Ring Labels to be displayed.

See Hiding and Activating Smith Chart Grid Lines.

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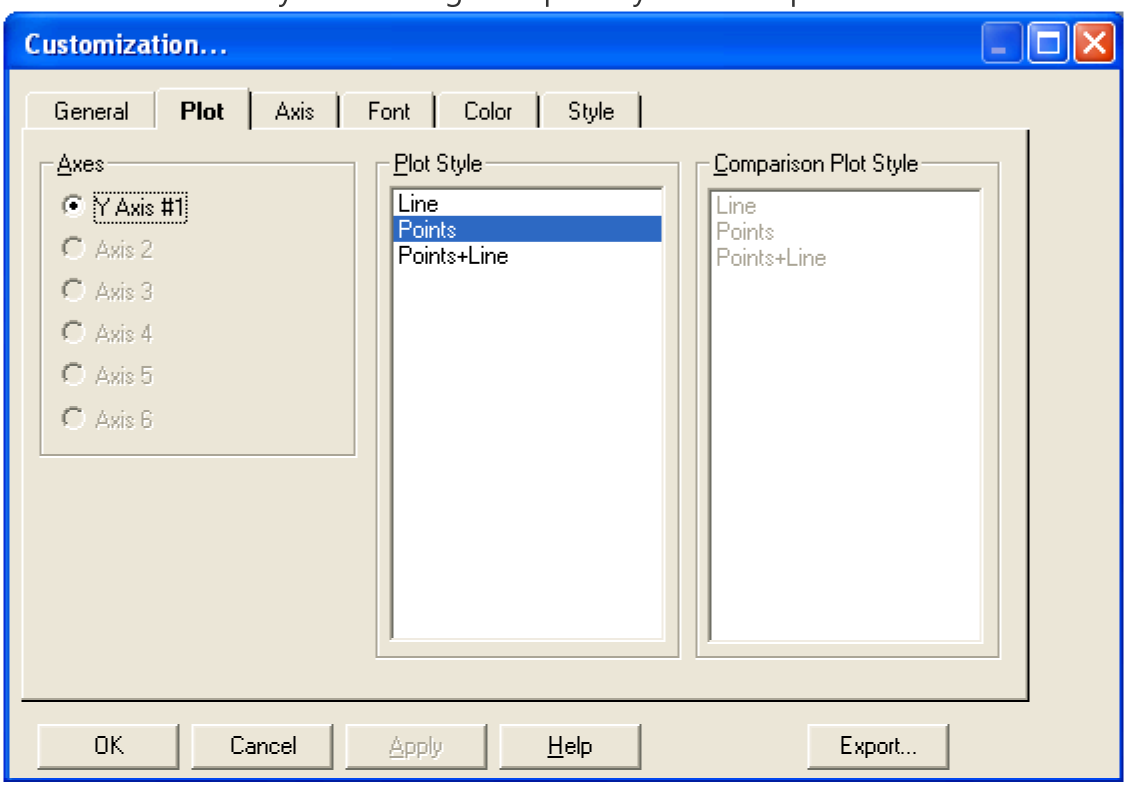
Smith Display : Smith Display Commands and Toolbars : Customization Dialog Box (Plot Tab)

Customization Dialog Box (Plot Tab)

The Customize>Customization Dialog command opens the multitabbed Customization dialog box that allows you to change the appearance of the Smith Display and plotted data.

The Plot tab allows you to change the plot style of data points on the Smith Chart.



Plot Tab Controls

Axes	Identifies the axis along which the data set points are plotted. This setting cannot be changed and is always set to Y axis #1.
Plot Style	Determines the style of the plotted data points on the display.

- Point: Each data point is represented by a symbol.
- Line: Each data point from the same data set is connected by a line.
- Points+Line: Plotted data is displayed in both the Point and Line plotting styles.

Comparison Plot Style	This setting is disabled and cannot be changed.

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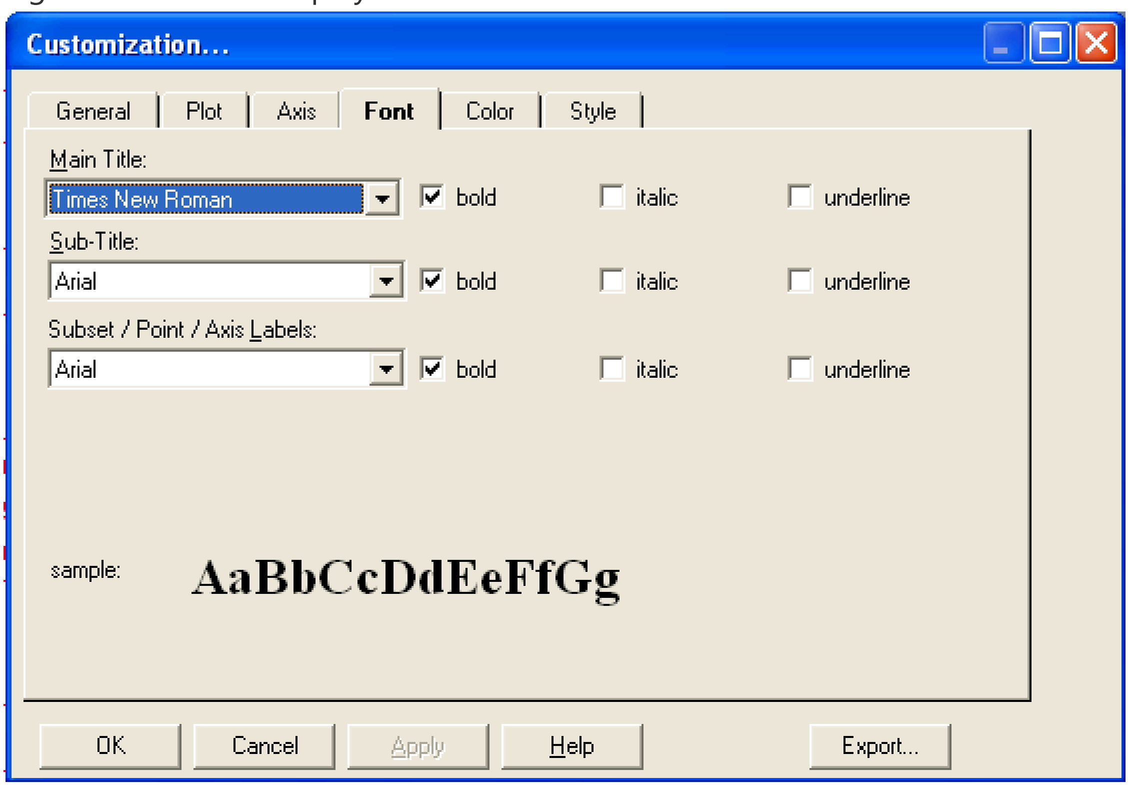
Smith Display : Smith Display Commands and Toolbars : Customization Dialog Box (Font Tab)

Customization Dialog Box (Font Tab)

The Customize>Customization Dialog command opens the multitabbed Customization dialog box that allows you to change the appearance of the Smith Display and plotted data.

The Font tab allows you to change the font styles and attributes for titles, reference labels, and legend text on the display.



Font Tab Controls

Main Title	Identifies the font style of the main title setting. The main title setting is not implemented, and changing the font setting does not affect the display.
Sub-Title	Identifies the font style of the sub-title setting. The sub-title setting is not implemented, and changing the font setting does not affect the display.
Subset / Point / Axis Labels	Identifies the font style of the reference labels and legend text displayed on the display.
bold	Applies the bold attribute to the text.
italic	Applies the italic attribute to the text.
underline	Applies the underline attribute to the text.
sample	Previews the font style as you make a font style selection in the Main Title, Sub-Title, or Subset/Point/Labels controls.

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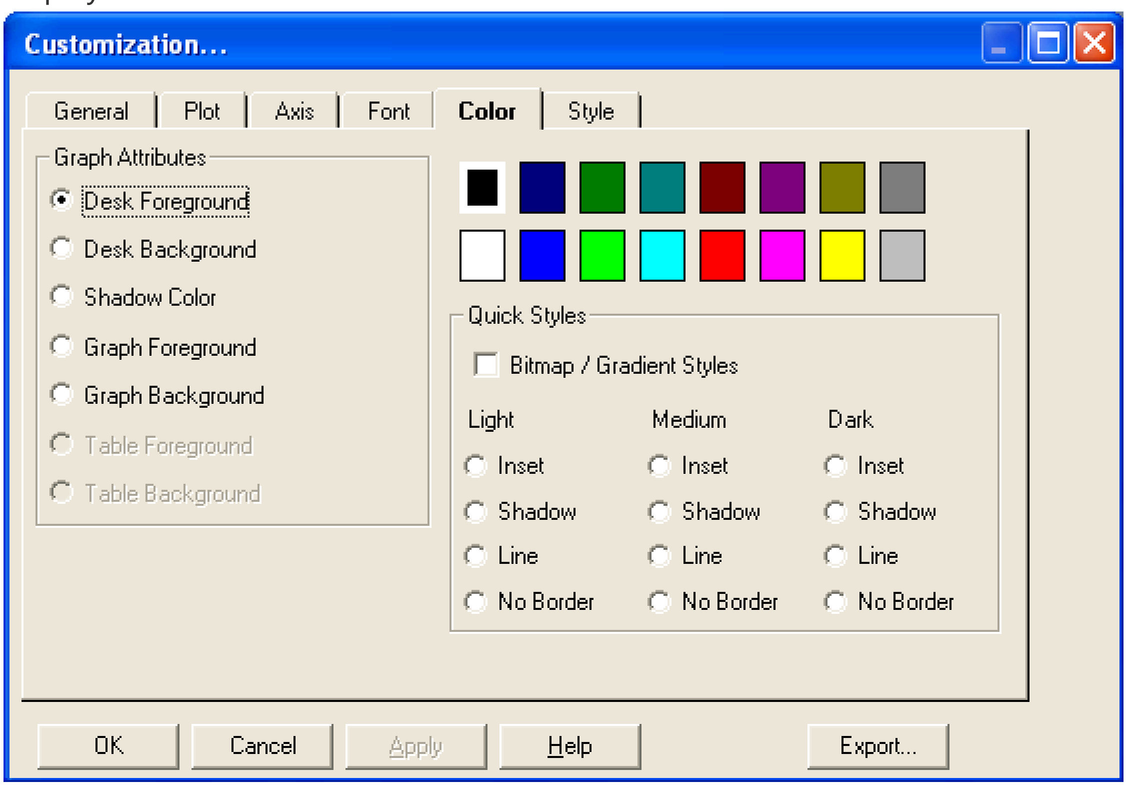
Smith Display : Smith Display Commands and Toolbars : Customization Dialog Box (Color Tab)

Customization Dialog Box (Color Tab)

The Customize>Customization Dialog command opens the multitabbed Customization dialog box that allows you to change the appearance of the Smith Display and plotted data.

The Color tab allows you to change the color of the foreground and background surfaces of the display window.



Color Tab Controls

Graph Attributes	Identifies the layer and structural components to which the selected color is applied.

- Desk Foreground: Applies the selected color to the text on the Desk Background. This includes the Data Set Summary title and the legend text.

- Desk Background: Applies the selected color to the window's background on which the Smith Chart's graph background is built (in particular, the background behind the data point table and the legend).

- Shadow Color: Applies the selected color to the rectangle that shadows the Smith Chart's Graph Background.

- Graph Foreground: Applies the selected color to the Smith Chart's grid lines and other structural components.

- Graph Background: Applies the selected color to the background on which the Smith Chart is immediately drawn.

Color Selection Grid	Displays the colors you can apply to graph attributes.
Quick Styles	Applies predefined viewing styles to the foreground and background window structures.

- Bitmap / Gradient Styles: Built-in background bitmaps or predefined gradients are used when selected; otherwise standard RGB colors are used for all backgrounds.

- Light Inset: Light colors with inset border
  Light Shadow: Light colors with shadow border
  Light Line: Light colors with line border
  Light No Border: Light colors with no border

- Medium Inset: Medium colors with inset border
  Medium Shadow: Medium colors with shadow border
  Medium Line: Medium colors with line border
  Medium No Border: Medium colors with no border

- Dark Inset: Dark colors with inset border
  Dark Shadow: Dark colors with shadow border
  Dark Line: Dark colors with line border
  Dark No Border: Dark colors with no border

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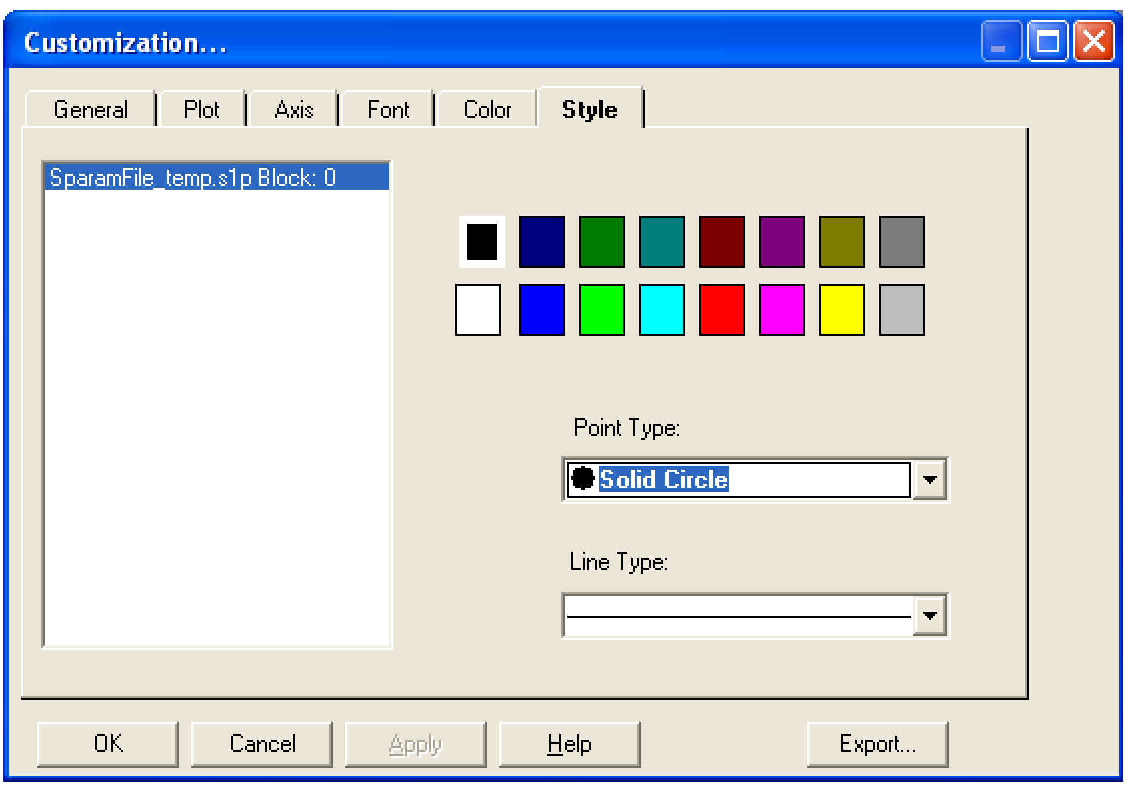
Smith Display : Smith Display Commands and Toolbars : Customization Dialog Box (Style Tab)

Customization Dialog Box (Style Tab)

The Customize>Customization Dialog command opens the multitabbed Customization dialog box that allows you to change the appearance of the Smith Display and plotted data.

The Style tab allows you to change the color, point type, and line type of data sets plotted on the Smith Chart.



Style Tab Controls

Data Set List	Identifies which data set style settings (color, point type, line type) are applied to.

Color Palette	Displays the colors you can apply to data point sets. Changing the color of a data set helps you to identify data points when multiple data sets are simultaneously plotted on the display. See Changing Data Point Type and Color.
Point Type	Identifies the graphical symbol that represents the plotted data of the selected data set when the style is set to Point or Point+Line. Changing the point type of a data set helps you to identify data points when multiple data sets are simultaneously plotted on the display. See Changing Data Point Type and Color.
Line Type	Identifies the style and thickness of the line that is used to connect data points of the same data set when the style is set to Line or Point+Line. Changing the line type of a data set helps you to identify data points when multiple data sets are simultaneously plotted on the display. See Changing Data Point Type and Color.

For more information, see Changing Data Point Type and Color.

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Smith Display : Smith Display Commands and Toolbars : Export Dialog Box

Export Dialog Box

The Customize>Export Dialog command opens the Exporting dialog box, which allows you to export an image of the Smith Display to the clipboard, save it to a file, or send it to a printer.

See also:

- Saving Data Sets and Plots
- Printing a Smith Display Plot

☑ **Note:**

Controls that appear in the Object Size portion of the Exporting dialog box depend on your selection in the Export and Export Destination. For more information, see Object Size in the table.

Export	Identifies the format in which the image of the display is to be exported. • Metafile: This format can be used for all destination selections (clipboard, file, and printer). The MetaFile format icon adapts to varying output resolutions. Since the image has textual attributes, however, do not stretch or squeeze it. If resizing, be sure to maintain the length to width ratio. • BMP: Bitmap (.bmp) is a graphical format that can only be used for the clipboard and file destinations. It is the most common image

export format. It uses a large amount of memory, however, and does not necessarily take advantage of high resolution printers.

- JPG: JPEG (.jpg) is a graphical format that can only be used for clipboard and file destinations. The JPEG format is compressed to take less memory, but it does not take full advantage of high resolution printers. This format is widely used on web pages. It is best suited for photographs because line art will be fuzzy due to compression. A better format is PNG.

- PNG: PNG is a graphical format that is used only for clipboard and file destinations. It is a compressed format that takes less memory, but it does not take advantage of high resolution printers. This format is often used on web pages, and it is most similar to the GIF format. It produces the best looking compressed images.

- Text/Data Only: This selection is not useful when using the Smith Display because the .s1p file already contains a text representation in the loaded .s1p file, and the text file that is generated by this setting is not in the .s1p format. This format can only be used for clipboard and file destinations. No Object Size options are available when this format is selected.

Export Destination	Identifies where the image of the display is exported.

- Clipboard: Copies an image of the display to the computer's clipboard. You can then paste the image into another application.

- File, Browse: Saves an image of the display to a file.

- Printer: Sends an image of the display to a printer producing a hardcopy.

Object Size	Specifies the physical dimensions of the exported image.

- No Specific Size: Only available when Metafile is selected in the Export region and either Clipboard or File is the selected destination.
- Full Page: Only available when Metafile is selected in the Export region and Printer is the selected destination.
- Pixels: Specifies the width and height size of the image in pixels. Only available when BMP, JPG, or PNG is selected in the Export region on the Exporting dialog box.
- Millimeters: Specifies the width and height size of the image in millimeters. Only available when Metafile is selected in the Export region on the Exporting dialog box.
- Inches: Specifies the width and height size of the image in inches. Only available when Metafile is selected in the Export region on the Exporting dialog box.
- Points: Specifies the width and height size of the image in points. Only available when Metafile is selected in the Export region on the Exporting dialog box.

Export	Sends the image to the clipboard or to a file. Only available when Clipboard or File is selected as the export destination. See Saving Data Sets and Plots.
Print	Sends the image to a printer. Only available when Printer is selected as the export destination. See Printing a Smith Display Plot.

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Smith Display : Starting the Smith Display

Starting the Smith Display

The Smith Display is typically used while debugging a test program. You can use it with the Microwave VNA Debug Display or the VBT Plot method in IG-XL or independently as a stand-alone application that can read s-parameter data files.

To start the Smith Display, follow one of these procedures:

- Starting the Smith Display in a Test Program
- Starting the Smith Display from TDE
- Starting the Smith Display as a Stand-Alone Executable

Starting the Smith Display in a Test Program

You can start the Smith Display while debugging s-parameter measurements by using the VBT Plot method for an S1PList or S2PList object in your program code. DataTool must be in debug mode.

To start the Smith Display from a test program in debug mode:

1. Add the Plot method to a VBT statement with the S1PList or S2PList measurement data object you want to plot on a display. See S1PList.Plot or S2PList.Plot.
2. Run the subroutine, function, or test program.

The S1P Smith Display (for an S1PList) or the S2P Smith Display (for an S2PList) appears.

Starting the Smith Display from TDE

The Smith Display is available in TDE. When you start Smith Display from TDE, it operates in stand-alone mode.

To start Smith Display from TDE:

- Select TDE>Tools>SmithDisplay.

The S1P Smith Display window appears in TDE.

Starting the Smith Display as a Stand-Alone Executable

To run the Smith Display as a stand-alone executable application, type the command smithdisplay in a Windows command prompt. Alternatively, you can open the smithdisplay.exe file in the \\Teradyne\IG-XL\V.xx.xx_yyyy\bin directory.

When opening the display from the command line, you can specify one or more filenames that automatically loads the data sets into the Smith Chart. For more information, see Loading Data Sets into the Smith Display's Stand-Alone Executable.

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Smith Display : Loading Data Sets into the Smith Display

Loading Data Sets into the Smith Display

To display s-parameter data on the Smith Display, the data must be arranged in the standard one-port s-parameter format within a text file. Although it is not required, the extension of the file containing the data set is typically .s1p. More than one data set can be stored in the same file. For more information, see Loading Multiple Data Sets into the Smith Display.

The following is an example of the standard one-port s-parameter format:

```
#  MHz         S  MA      R  50.0
! frequency    s11.mag    s11.arg
10.000000      0.157161    45.8500
20.000000      0.242265    13.7002
30.000000      0.252887    3.5127
40.000000      0.252680   -2.7557
```

Typically, you read data into the display in one of the following ways:

- Loading Data Sets Using the Open Command
- Loading Data Sets from VBT
- Loading Data Sets into the Smith Display's Stand-Alone Executable

You can also load multiple data sets to the display at the same time:

- Loading Multiple Data Sets into the Smith Display

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Smith Display : Loading Data Sets into the Smith Display : Loading Data Sets Using the Open Command

Loading Data Sets Using the Open Command

When you start the Smith Display as a stand-alone executable (without a file name), the display has no data loaded. Use the Open command to open a data file and load a data set to the display.

To load a data set to the Smith Display:

1. Select File>Open.

The Open dialog box appears and prompts you to enter or browse for a file name.

2. Select a file to plot on the Smith Chart.

If the file contains exactly one s-parameter data set, selecting the file automatically plots the data set on the Smith chart.

If the selected file contains more than one data set, a dialog box appears and prompts you to select one, some, or all of these data sets for plotting. To select multiple data sets, hold down the Shift key and click the desired data set names. For more information, see Loading Multiple Data Sets into the Smith Display.

For information on viewing individual data sets once multiple data sets are loaded, see Displaying Data Point Information Using the Data Sets Command.

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Smith Display : Loading Data Sets into the Smith Display : Loading Data Sets from VBT

Loading Data Sets from VBT

Data sets in S1PList or S2PList objects are automatically loaded and plotted if a statement with the Plot method is in the VBT code and DataTool is in debug mode.

See Starting the Smith Display in a Test Program.

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Smith Display : Loading Data Sets into the Smith Display : Loading Data Sets into the Smith Display’s Stand-Alone Executable

Loading Data Sets into the Smith Display’s Stand-Alone Executable

As described in Starting the Smith Display as a Stand-Alone Executable, you can start the Smith Display as a stand-alone executable by specifying smithdisplay at the Windows Command prompt. Specifying this command opens an empty Smith Chart (no plotted data), and you will need to use the Open command to load data to the display. To avoid this extra step, you can specify the file name immediately after the smithdisplay command.

For example to load one file, type:

smithdisplay a.s1p

To load multiple files, type:

smithdisplay a.s1p, b.s1p, c.s1p

☑ **Note:**

Directory path names and wildcards are supported, and you can specify multiple files separated by a space, tab, or comma.

For more information, see Loading Data Sets Using the Open Command.

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Smith Display : Loading Data Sets into the Smith Display : Loading Multiple Data Sets into the Smith Display

Loading Multiple Data Sets into the Smith Display
To plot multiple data sets simultaneously on the Smith Chart, load a file containing multiple data sets or load single data sets consecutively.

- Loading Multiple Data Sets From One File
- Loading Multiple Data Sets From Several Individual Files

☑ **Note:**
Once a data set is plotted on the Smith Chart it cannot be removed, nor can the Smith Chart be reset. You can, however, isolate data sets by hiding individual data sets using the S-Parameter Data Sets dialog box (see Displaying Data Point Information Using the Data Sets Command) or reload the Smith Display.
Loading Multiple Data Sets From One File
A single file can contain multiple data sets. When you load a file that contains multiple data sets into the Smith Display (using the Open command), a dialog box appears listing all data sets contained in the file. You can load only one data set by clicking the name of a single set, or you can select multiple sets by holding down the Shift key and clicking the desired sets. Once you select the desired sets, click OK and the data is automatically plotted on the Smith Chart. Note that data sets are superimposed on each other and are automatically distinguished by color. To make individual data set plots easier to distinguish, you can also change the point styles.
For more information, see Changing Data Point Type and Color.
Loading Multiple Data Sets From Several Individual Files

Multiple data sets can be superimposed on the Smith Chart using the Open command repeatedly or by supplying multiple file names to the smithdisplay command.

☑ **Note:**

If you load a file with multiple data sets at the command line, all data sets in the file are automatically plotted.

For more information, see Loading Data Sets Using the Open Command.

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Smith Display : Viewing Data Point Information

Viewing Data Point Information

The Smith Display is useful for debugging test programs because it allows you to verify your expected RF signal measurements. Note that the Smith Display is read-only and that you cannot alter plotted data in any way. Still, you can collect useful debugging information by examining the location and characteristics of data points plotted on the Smith Chart.

The following sections describe ways the display allows you to examine data point information:

- Displaying Data Point Information in the Data Point Table
- Displaying Data Point Information Using Markers
- Displaying Data Point Information Using the Data Sets Command
- Zooming In and Out of a Data Point on the Smith Display

☑ **Note:**

Once a data set is plotted on the Smith Chart, you cannot remove it, nor can you reset the Smith Chart. However, you can hide individual data sets using the S-Parameter Data Sets dialog box (see Hiding and Showing Data Sets) or start a new Smith Display session.

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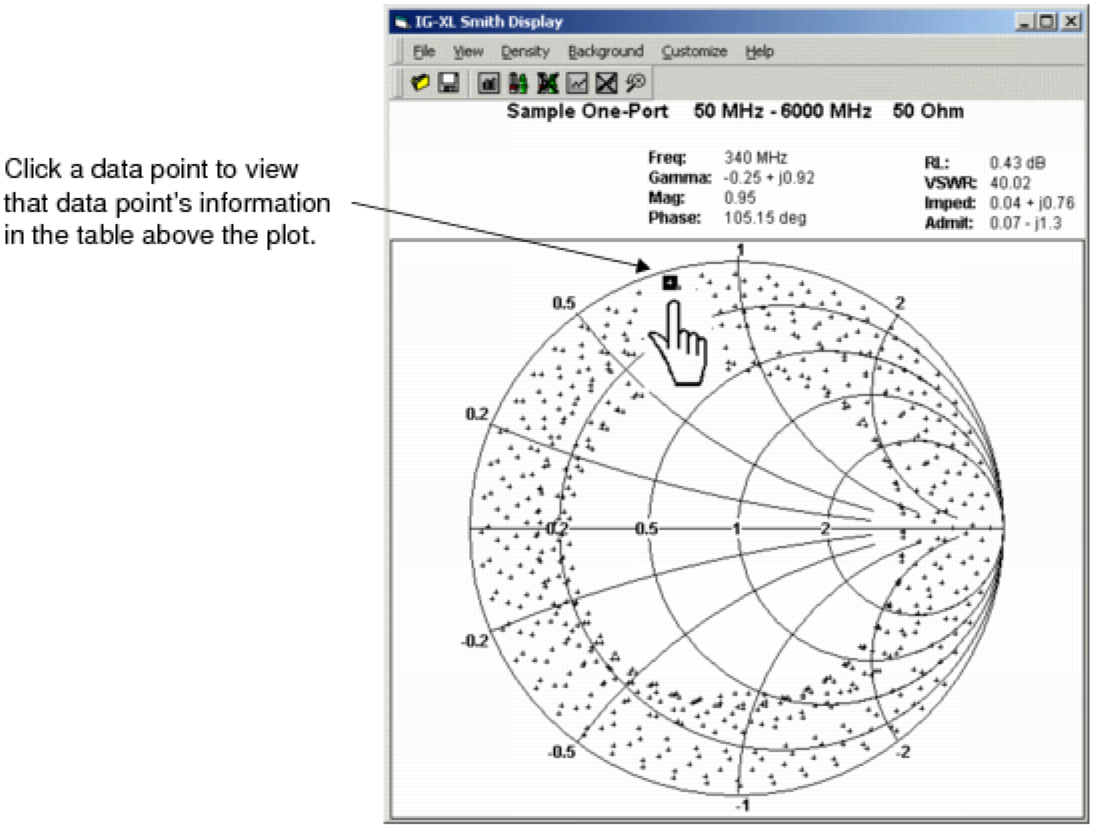
Smith Display : Viewing Data Point Information : Displaying Data Point Information in the Data Point Table

Displaying Data Point Information in the Data Point Table

When you load a data set into the Smith Display, it is plotted on the chart, and information about the data set (start and stop frequency, and characteristic impedance) is shown in the title at the top of the display window.

While the title shows data set information, the Data Point Table displays information about a particular data point. By default, the Data Point Table is visible at the top of the display window. Using the Show Data On  and Show Data Off  commands, you can display or hide the Data Point Table from the display.

The figure shows that when you click or mouse over a plotted data point, information about that data point is displayed in the Data Point Table. The Data Point Table shows the frequency, gamma, magnitude, phase, return loss (RL), voltage standing wave ratio (VSWR), impedance, and admittance for the selected data point.



Viewing Data Point Information

☑ **Note:**

The Smith Display is read-only. Therefore, you cannot edit or change the values of any plotted data.

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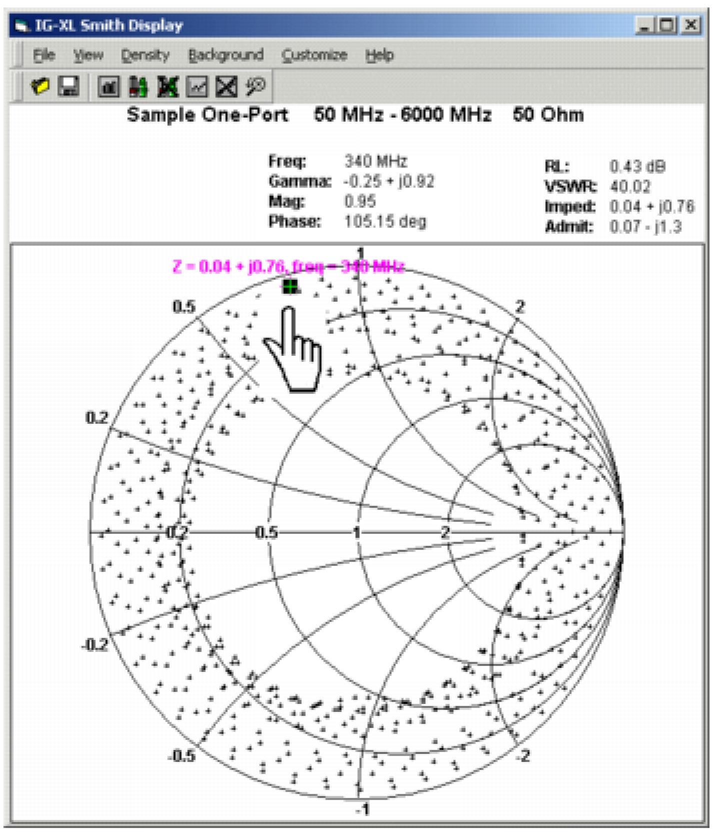
Smith Display : Viewing Data Point Information : Displaying Data Point Information Using Markers

Displaying Data Point Information Using Markers

The Smith Display's markers allow you to view the frequency associated with a data point and the associated impedance Z or admittance Y, according to the selected chart background.

By default, markers are turned off (Markers Off), but you can activate them using the Markers On command. The figure shows that when markers are turned on, moving the mouse pointer over a data point causes a marker to float over it displaying a brief string of information about that data point. Click the data point and the marker text remains displayed next to the data point as a reference.



Data Point Marker

To clear a marker from a data point, click the point again.

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Smith Display : Viewing Data Point Information : Displaying Data Point Information Using the Data Sets Command

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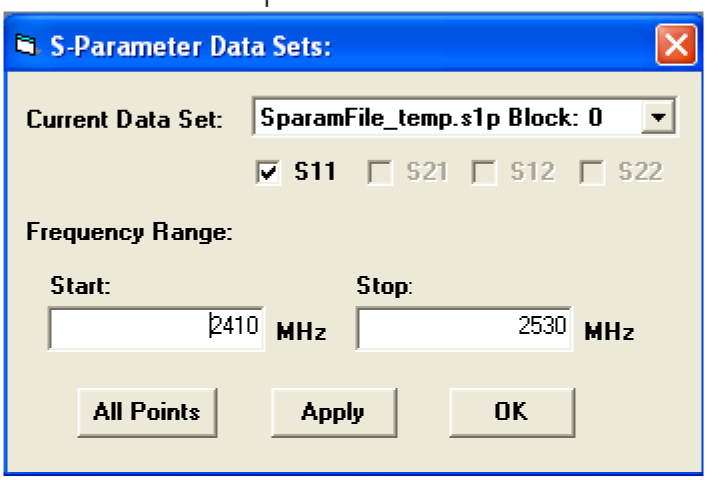
Displaying Data Point Information Using the Data Sets Command

The Data Sets command opens the S-Parameter Data Sets dialog box, which allows you to show or hide selected s-parameter data sets and set the frequency range for a selected data set.



S-Parameter Data Sets

Hiding and Showing Data Sets

Once a data set is plotted on the Smith Chart, you cannot remove it, nor can you reset the Smith Chart. However, you can hide selected data sets using the S-Parameter Data Sets dialog box from the Data Sets command.

To hide a selected data set:

1.	Select the Data Sets command.

The S-Parameter Data Sets dialog box appears.

2. Select the name of the data set that is to be hidden from the Current Data Set drop-down menu.

3. Deselect the S11 check box, or the check box for another term, and click OK.

To show a hidden data set:

1. Select the Data Sets  command.

The S-Parameter Data Sets dialog box appears.

2. Select the name of the data set that is to be reinstated from the Current Data Set drop-down menu.

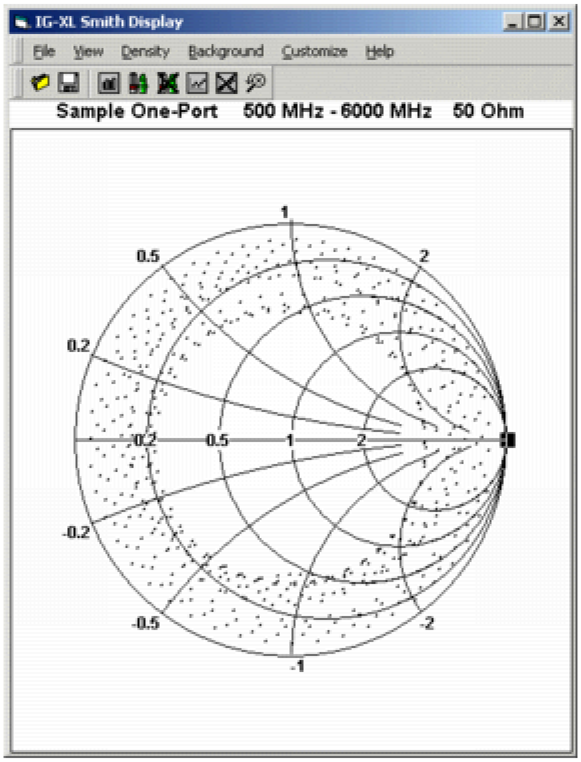
3. Select the S11 check box, or the check box for another term, and click OK.

The selected data set is again plotted on the Smith Chart.

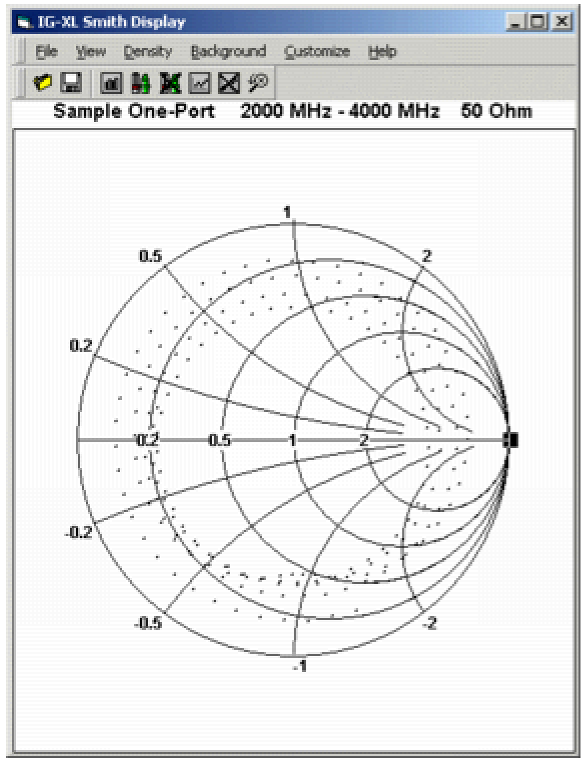
Viewing Data Points Plotted Within a Specific Frequency Range

The Frequency Range portion of the S-Parameter Data Sets dialog box contains two fields: Start and Stop. When a data set is loaded into the Smith Display, the beginning and ending frequency limits of the plotted data set appears automatically in these two fields.

The figure shows that by changing the Start and Stop fields you can instruct the display to show only the data points in a specific frequency range.



Frequency Range
Start: 500 MHz
Stop: 6000 MHz



Frequency Range
Start: 2000 MHz
Stop: 4000 MHz

Viewing Data Points Plotted Within a Specific Frequency Range

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Smith Display : Viewing Data Point Information : Zooming In and Out of a Data Point on the Smith Display

Zooming In and Out of a Data Point on the Smith Display

You can zoom in on a selected area of the Smith Chart to have a closer look at a particular set of data points. To zoom in, hold Shift while clicking and dragging the left mouse button to define the area for zooming. Release the mouse button once the zoom area is defined and the display zooms in on that region.

To undo the zoom, press  or Z.

☑ **Note:**

Currently only one level of zooming is supported. This means you can either have a zoomed-in view of a particular area or a complete view of the Smith Chart.

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Smith Display : Changing the Appearance of the Smith Display

Changing the Appearance of the Smith Display

You can change many of the Smith Display's visual attributes to customize the presentation of plotted s-parameter data.

The following sections describe how you can change the appearance settings:

- Changing Data Set Plot Style
- Changing Data Point Type and Color
- Changing the Background Between Impedance and Admittance
- Hiding and Activating Smith Chart Grid Lines
- Changing the Ring Density
- Showing and Hiding the Smith Display Legend

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Smith Display : Changing the Appearance of the Smith Display : Changing Data Set Plot Style

Changing Data Set Plot Style

The Smith Display offers three visual styles for plotting s-parameter data points on the Smith Chart:

Points	Places an individual symbol at the exact data point on the Smith Chart.
Line	Connects a line between each data point.
Points+Line	This style is a combination of Points and Line.

When multiple data sets are loaded and plotted on the display, all data sets have the same style. To help distinguish between data sets, you can change the point and line types or the color of the data sets to be unique from one another.
To change data set plot style:

1.	Select Customize>Customization Dialog.

The Customization dialog box appears.

2.	Select the Plot tab on the Customization dialog box.

3.	Select the desired data plot style (Line, Points, or Points+Line) from the Plot Style box.

4.	Click Apply.

The plot style is immediately previewed on the display.

5.	Click OK to complete the plot style change.

Related Topics

- Customization Dialog Box (Plot Tab)
- Changing Data Point Type and Color

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Smith Display : Changing the Appearance of the Smith Display : Changing Data Point Type and Color

Changing Data Point Type and Color

Multiple data sets that are plotted simultaneously on the Smith Chart all have the same style applied to them when loaded. To change the appearance of these data points (point/line type and color) so that you can distinguish between data sets, follow this procedure:

1.	Select Customize>Customization Dialog.

The Customization dialog box appears.

2.	Select the Style tab on the Customization dialog box.

3.	Select the desired data set in the Data Set List box.

The color palette, point type, and line controls show the current settings for the data set.

4.	Select the desired color from the color palette.

As you select different colors, the change is previewed in the display. This preview feature, however, does not apply the change.

5.	Select the desired point type from Point Type drop-down list.

As you select different point type, the change is previewed in the display. This preview feature, however, does not apply the change.

6.	Select the desired line type from Line Type drop-down list.

As you select different point type, the change is previewed in the display. This preview feature, however, does not apply the change.

7.	Click OK.

The data point and line changes are now applied to the display.

For more information, see:

- Customization Dialog Box (Style Tab)
- Changing Data Set Plot Style

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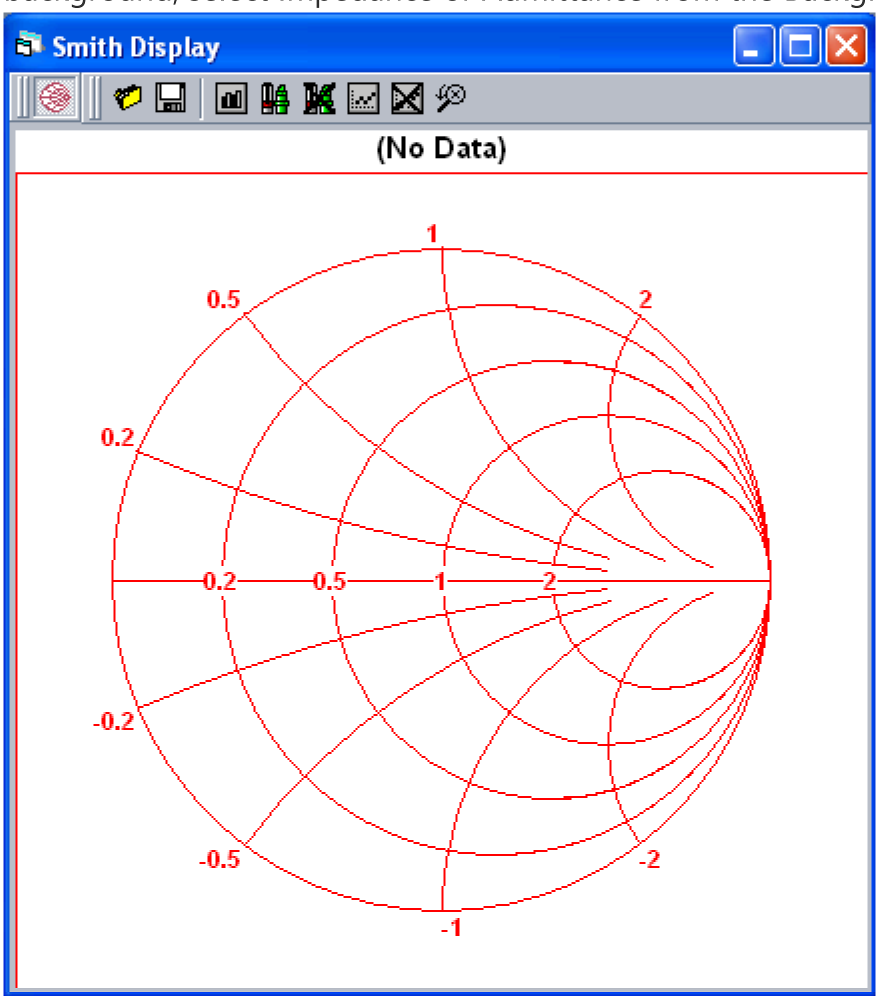
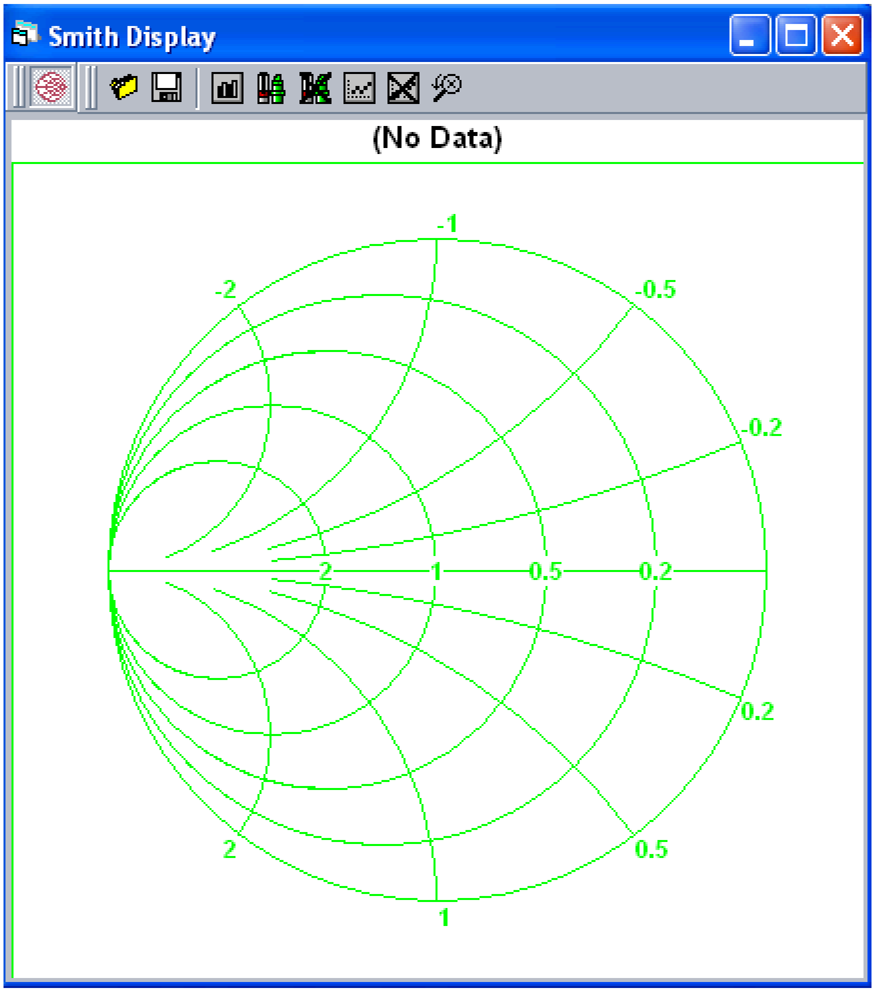
Smith Display : Changing the Appearance of the Smith Display : Changing the Background Between Impedance and Admittance

Changing the Background Between Impedance and Admittance

The figure shows that the Smith Chart background can be Impedance or Admittance. To change the background, select Impedance or Admittance from the Background menu.



Smith Chart Background, Impedance Format



Smith Chart Background, Admittance Format

See The Smith Chart for an explanation of how the Smith Chart appearance changes depending on the background.

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Smith Display : Changing the Appearance of the Smith Display : Hiding and Activating Smith Chart Grid Lines

Hiding and Activating Smith Chart Grid Lines

The Smith Chart can become crowded when multiple data sets are plotted. To improve the visibility of a plotted trace, you can hide the Smith Chart rings.

To show or hide the Smith Chart’s grid lines:

1.	Select Customize>Customization Dialog.

The Customization dialog box appears.

2.	Select the General tab on the Customization dialog box.

3.	Select one of the following settings from the Grid Lines control:

–	R: Shows the horizontal (R) rings only. Hides the vertical (X) rings.

–	X: Shows the vertical (X) rings only. Hides the horizontal (R) rings.

–	None: Shows no grid lines at all, nor the center horizontal line.

4.	Click Apply.

The grid line changes are immediately previewed on the display.

5.	Click OK.

For more information, see:

- Customization Dialog Box (General Tab)
- Changing the Ring Density

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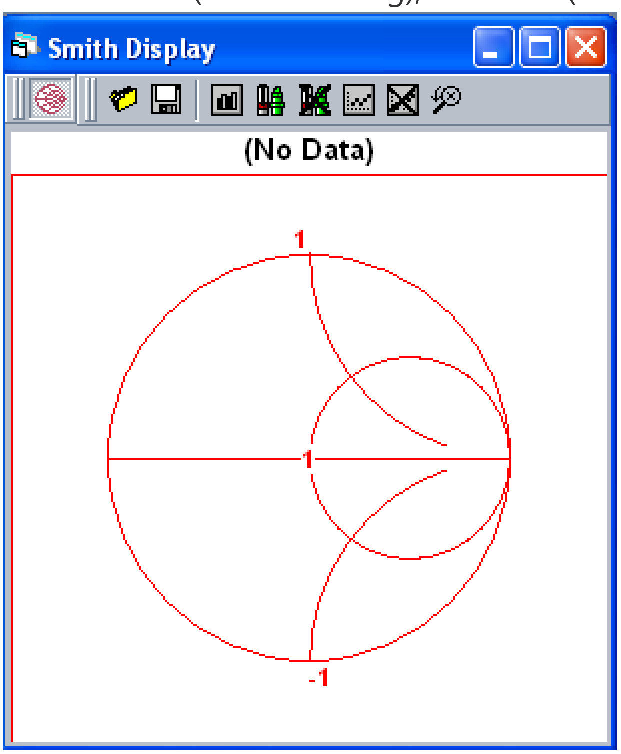
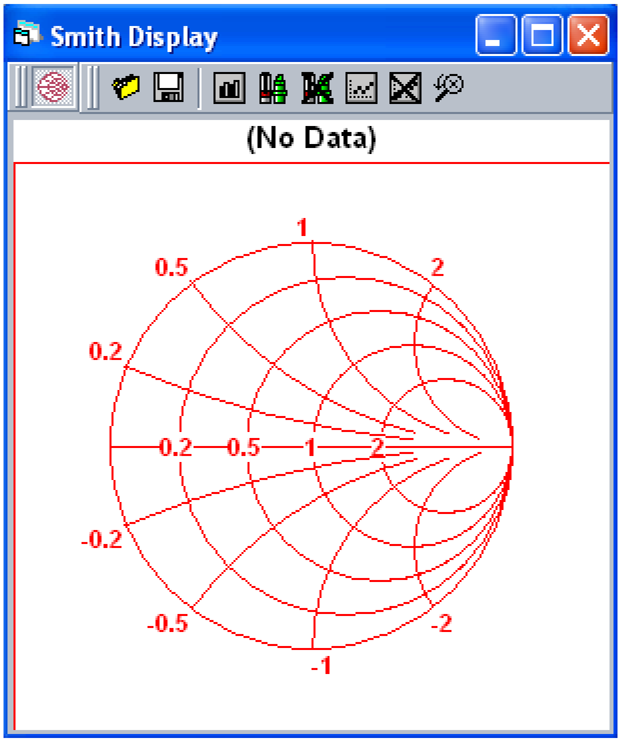
Smith Display : Changing the Appearance of the Smith Display : Changing the Ring Density

Changing the Ring Density

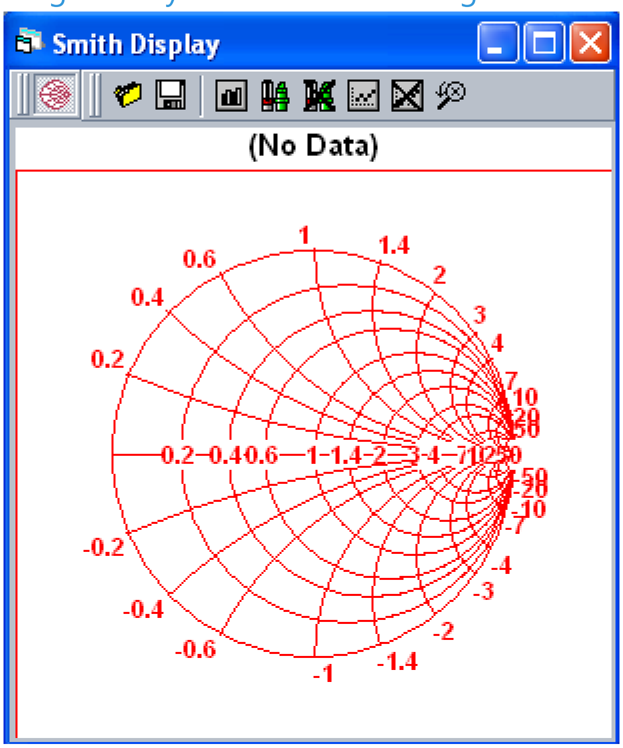
You can set a Smith Chart's ring density to display more or less reference lines. Use the Ring Density command to change the ring density to one of the following settings: Simple (less lines), Intermediate (default setting), or Dense (more lines)



Ring Density Simple Setting



Ring Density Intermediate Setting



Ring Density Dense Setting

For more information, see The Smith Chart.

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Smith Display : Changing the Appearance of the Smith Display : Showing and Hiding the Smith Display Legend

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Showing and Hiding the Smith Display Legend

To help you quickly identify data set measurements on a multiple data set plot, the Smith Display provides a legend that identifies and associates data sets with the point type symbol that appears on the Smith Chart.

You can expand and collapse the legend in the lower part of the display window. Use the Show Legend command to display the legend and the Legend Off  command to hide the legend.

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Smith Display : Saving Data Sets and Plots

Saving Data Sets and Plots

The Smith Display allows you to save a copy of one or more data sets to a file. Note, however, that since the display cannot change data, you need not save the original file. Instead, you can use the save functionality to save a data set to a file, which is different from the original file, or to combine multiple data sets from different files into one file.

To save data sets to a new file:

1. Select the File>Save command.

The Select Blocks To Save dialog box appears.

2. Select one, multiple, or all data sets to be saved in the new file and click OK.

The Save As dialog box appears.

3. Specify the file name and location for the new file and click Save.

You can also extract and save an image of the Smith Display for future reference. To save an image of a Smith Display plot:

1. Select the Export Dialog command.

The Exporting dialog box appears.

2. Select the file type format from the Export section of the Export dialog box. You can select bitmap (.bmp), JPEG (.jpg), or PNG (.png) formats.

3. Select File as the Export Destination.

The Browse button becomes active.

4. Click the Browse button.

The Save As dialog box appears.

5. Navigate to the directory location where you want to save the file, specify the file's name, and click Save.

The Save As dialog box closes.

6. Specify the desired dimensions of the graphical representation of the display in the Object Size portion of the Exporting dialog box.

7. Click the Export button in the Exporting dialog box.

The display plot is saved. You can open it using a graphics viewer/editor application.
For more information, see:

- Export Dialog Box
- Printing a Smith Display Plot

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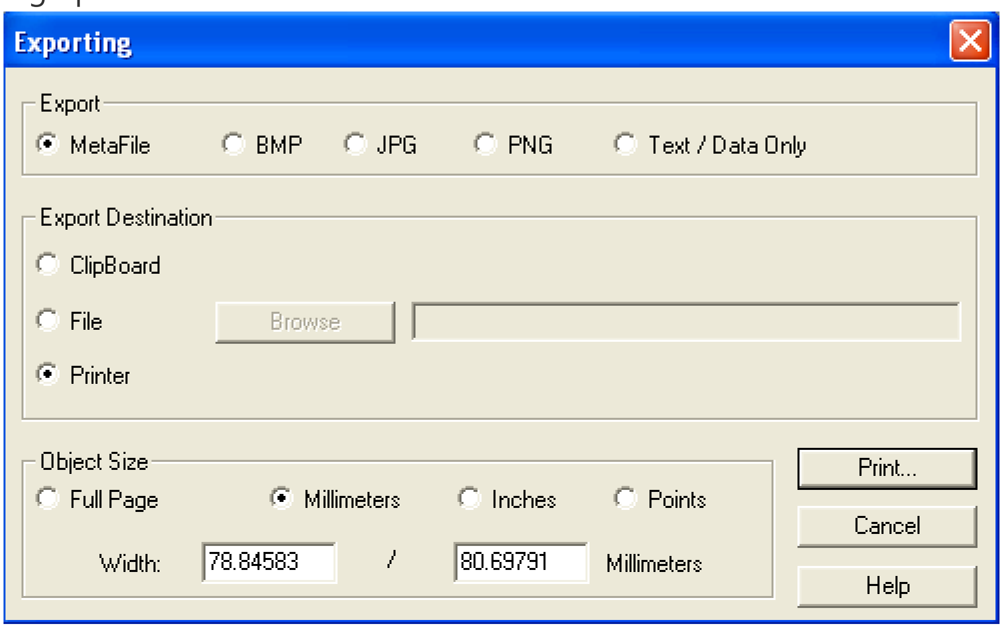
Smith Display : Printing a Smith Display Plot

Printing a Smith Display Plot

Once you plot and analyze a data set on the Smith Display, you can print a static version of the plot for future reference. The figure shows that by using the Export control, you can convert the plot to a graphics or text format and then either save it to a file or send it to a printer.



Export Plot to Printer

To print a Smith Display plot:

1. Select the Export Dialog command.

The Exporting dialog box appears.

2. Select the file type format from the Export section of the Export dialog box. You can select bitmap (.bmp), JPEG (.jpg), or PNG (.png) formats.

3.	Select Printer as the Export Destination.

4.	Specify the desired dimensions of the graphical representation of the display in the Object Size portion of the Exporting dialog box. Typically, Full Page is used.

5.	Click the Print button in the Exporting dialog box.

The Printing dialog box appears.

6.	Specify the Printer destination and the Printing Style.

7.	Click OK.

The display plot is sent to the printer. You now have a paper copy of the plot graphic.

For more information, see:

- Export Dialog Box
- Saving Data Sets and Plots

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Smith Display

Smith Display

The IG-XL Smith Display tool plots one-port or two-port s-parameter measurement data on a Smith chart. The MW VNA instrument, or another network analyzer, performs the measurements and creates the s-parameter data. The Smith Display accepts data in S1PList, S2PList, S1PExList, or S2PExList objects or data files and displays data sets in two different display windows:

S1P Smith Display	This is the full-featured display, with a Smith chart and data evaluation features. It shows a plot for a single s-parameter term at a time, such as s11. Since two-port s-parameter terms are stored as S1P objects in an S2PList, it can also show s12, s21, and s22 when started from the S2P Smith Display.
S2P Smith Display	The S2P display shows all four s-parameter terms of a two-port measurement and provides easy access to the S1P Smith Display (Smith chart) or the WaveScope (samples vs. time plot) for each term.

During interactive debugging, you can start the Smith Display in IG-XL debug mode from VBT (S1PList.Plot, S2PList.Plot, S1PExList.Plot, or S2PExList.Plot methods).

For off-line analysis of s-parameter data, you can start the Smith Display as a stand-alone tool from either the TDE Tools menu or the Windows command prompt, and then load individual s-parameter data files.

The following sections describe the Smith Display and how to use it:

- The Smith Chart

- S1P Smith Display
- S2P Smith Display
- Smith Display Commands and Toolbars
- Starting the Smith Display
- Loading Data Sets into the Smith Display
- Viewing Data Point Information
- Changing the Appearance of the Smith Display
- Saving Data Sets and Plots
- Printing a Smith Display Plot

For more information on VNA instruments, see About the UltraWave 12G VNA

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